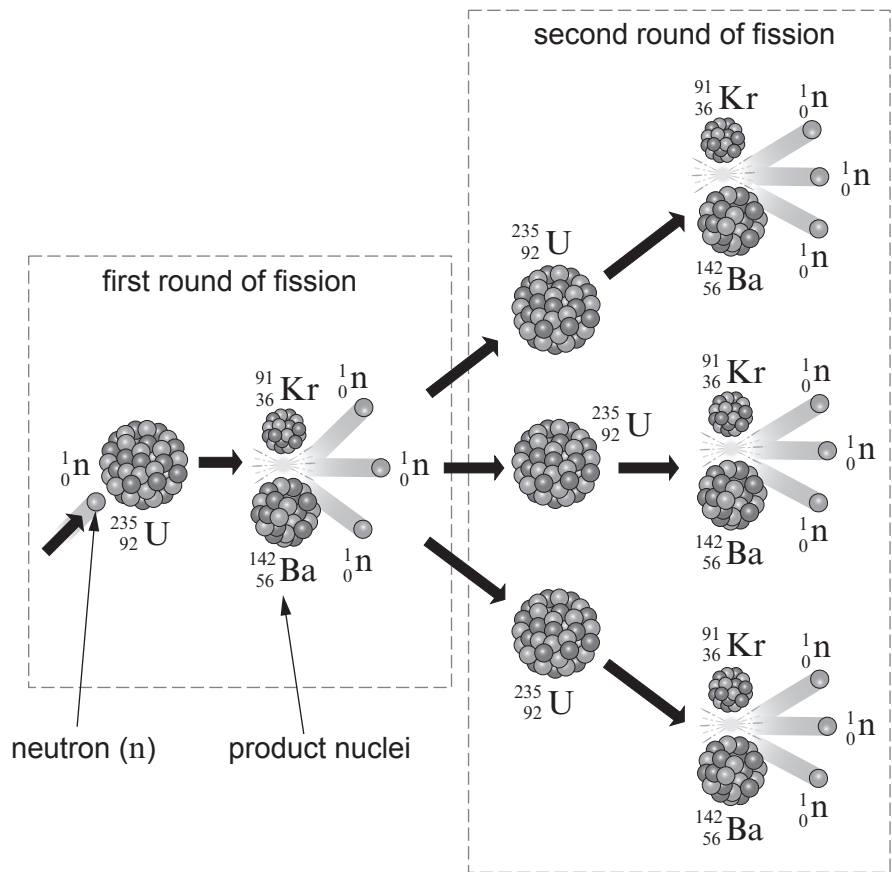


5. The diagram shows one possible chain reaction in the fission of uranium-235 ($^{235}_{92}\text{U}$) by a single neutron (^1_0n).
The product nuclei, krypton (Kr) and barium (Ba), are radioactive.



During the first round of fission, 2 product nuclei are created and 3 neutrons are released.

- (a) Tick (✓) the **three** boxes next to the correct statements below.

[3]

One neutron splits to create three other neutrons.

☐

The proton numbers of barium and krypton add to give the proton number of uranium-235.

☐

Nuclear fission takes in energy from the nucleus.

☐

During the second round of fission, another 9 neutrons are released.

☐

During the second round of fission, another 6 product nuclei are created.

☐

An atom of uranium-235 has 92 neutrons in its nucleus.

☐

Examiner
only

(b) Use words from the box to complete the following sentences about a nuclear reactor. Each word may be used once, more than once or not at all.

absorb	one	two	three	fuel	moderator
emit	concrete	reactor	control		

- (i) Neutrons are slowed down by the so uranium nuclei
can them. [2]
- (ii) rods absorb fission neutrons so only
..... of these neutrons goes on to produce further fission. [2]
- (iii) In an emergency, rods are dropped fully into the
nuclear [2]

9

